Database Management: Normalization

What Is Normalization

Normalization is the process of organizing the columns and tables of a relational database to reduce data redundancy and avoid undesirable characteristics like insertion, update, and deletion anomalies. It works by decomposing larger tables into smaller, well-structured tables and defining relationships between them using foreign keys.

First Normal Form (1NF)

A table is in First Normal Form if every column contains only atomic, indivisible values, and each record is unique. This means a single column should not store multiple values, such as a comma-separated list of phone numbers. Instead, related repeating values should be moved into a separate table.

Second Normal Form (2NF)

A table is in Second Normal Form if it is already in 1NF and every non-key column depends on the entire primary key, not just part of it. This rule only becomes relevant when a table has a composite primary key made of more than one column. Partial dependency on only part of the composite key violates 2NF.

Third Normal Form (3NF)

A table is in Third Normal Form if it is already in 2NF and no non-key column depends on another non-key column. In other words, all non-key columns must depend only on the primary key, and not transitively on some other non-key column. This eliminates what is known as a transitive dependency.

Most practical relational database designs aim for Third Normal Form, since it offers a good balance between eliminating redundancy and keeping queries reasonably simple. Higher normal forms, such as Boyce-Codd Normal Form (BCNF) and Fourth Normal Form (4NF), exist for edge cases involving more complex dependency structures.

Tradeoffs of Normalization

While normalization reduces redundancy and improves data integrity, it can increase the number of joins needed to answer a query, which may hurt read performance. For this reason, some systems intentionally denormalize parts of their schema, especially in read-heavy analytical workloads, trading some redundancy for faster queries.